



Prevalence, antimicrobial resistance and virulence genes of *Salmonella* serovars isolated from humans and animals

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Abstract

We investigated the prevalence, antimicrobial susceptibility, antimicrobial resistance and virulence genes of *Salmonella* isolates recovered from humans and different species of animals. Out of 1231 samples, 88 (7.15%) *Salmonella* isolates were obtained, among which 21 (23.86%) belonged to *Salmonella enterica* subsp. *enterica* sero var. Weltevreden, 22 (25%) to *S. Enteritidis*, 16 (18.2%) to *S. Typhi* and 14 (15.9%) to *S. Newport*; 7 (7.95%) isolates were untypable. Among the 88 isolates, 65.90% showed resistance to gentamicin, 61.36% to tetracycline, 61.18% to cefotaxime, 48.86% to trimethoprim, 45.45% to ampicillin, 11.36% to ceftriaxone, 10.22% to chloramphenicol and 7.95% each to ciprofloxacin and cefepime. Most of the isolates were susceptible, with a low MIC (≤ 0.25 µg/ml) value, to cefepime, cefotaxime, ciprofloxacin, ceftriaxone and co-trimoxazole and with a moderate MIC (0.5–4 µg/ml) to ampicillin, tetracycline, gentamicin and chloramphenicol. The resistance genes *bla*_{TEM}, *tetA* and *dfrA12* were most prevalent, irrespective of the host of origin of the isolates. While *invA* was used for molecular detection of *Salmonella*, other virulence genes, viz. *sipA*, *sipB*, *sipC*, *stn* and *pagN*, were also detected in all *Salmonella* isolates. A total of 38.64% isolates were multidrug-resistant (MDR), and various virulence genes were present among the isolated serovars. This study highlights the importance of continuous monitoring and surveillance for pathogenic *Salmonella* and their potential risks to both humans and animals.

Keywords Antimicrobial resistance · Prevalence · *Salmonella* · Virulence gene

Introduction

Bacteria of the genus *Salmonella* are found in the intestinal tract of numerous animals, including cattle, pigs, horses, other mammals, reptiles, amphibians and poultry (e.g., chickens, ducks, geese and turkeys) and can cause infections (Hale et al. 2012). In humans, *Salmonella* infection typically manifests as acute gastroenteritis that develops 12–72 h after exposure. Children, persons >65 years and immuno-compromised people are at greater risk of serious complications, including septicaemia, joint or brain infections and death (Giannella 1996). The genus *Salmonella* consists of two species, namely

Salmonella bongori and *Salmonella enterica* (Issenhuth-Jeanjean et al. 2014). *Salmonella bongori* is also termed the “*Salmonella* of lizards” as it is mostly found in reptiles, causing diarrhoea. *Salmonella enterica* is predominantly found in humans and animals, causing severe diarrhoea and fever. The species *S. enterica* contains six subspecies, namely *enterica*, *salamae*, *arizonae*, *houstonae*, *indica* and *diarizonae*, all with different serotypes. So far, 2659 serotypes of *S. enterica* have been reported (Monte and Sellera 2020). Of these, *S. enterica* subsp. *enterica* has the highest number of serotypes (1586) (Issenhuth-Jeanjean et al. 2014). Although all serotypes must be considered potential human pathogens, only a limited number of serotypes can cause infections in humans and animals. Most of the enteric diseases in humans and animals with severe diarrhoea and fever are caused by different serotypes of *S. enterica* subsp. *enterica*. This subspecies is one of the leading causes of zoonotic food-borne disease worldwide (Voetsch et al. 2004). It is estimated that in the USA, in 2019, there were 2,12,500 and 4100 cases of infections due to drug-resistant

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non-typhoidal *Salmonella* and drug-resistant *S. typhi*, respectively (CDC Report 2019).

The Antimicrobial Resistance Surveillance and Research Network (AMRSN) of India has categorised *Salmonella* (typhoidal and non-typhoidal) into enteric fever pathogens and diarrheagenic bacterial organism groups out of a total of six groups selected by the Indian Council for Medical Research for developing a comprehensive plan for antimicrobial resistance (AMR) surveillance in India (Walia et al. 2019). *Salmonella* species, specifically those that are fluoroquinolone resistant, are categorised as “high priority” species by World Health Organization (WHO Report 2017). Multidrug-resistant (MDR) *Salmonella* carrying several classes of virulence genes have been detected in duck meat in China (Chen et al. 2020; Wu et al. 2021; Yingyu et al. 2021). Outbreaks of *S. Typhi* recently occurred in Bangladesh (Tanmoy et al. 2018) and Pakistan (Klemm et al. 2018), with the presence of extensively drug-resistant (XDR) strains resistant to three first-line drugs (chloramphenicol, ampicillin and trimethoprim-sulfamethoxazole) as well as fluoroquinolones and third-generation cephalosporins (Klemm et al. 2018). *Salmonella* Typhimurium showed a high frequency of occurrence in poultry (41.40%) and humans (43%), and *S. Weltevreden* was found to be of zoonotic significance in India and has been recorded as one of the five most frequently isolated serovars (Kumar et al. 2009). In Assam, isolation of *S. Weltevreden*, *S. Choleraesuis*, *S. Paratyphi B* and *S. Typhimurium* from pigs (Rajkhowa et al. 2018), and *S. Enteritidis*, *S. Gallinarum*, *S. Typhimurium*, *S. Newport* and *S. Indiana* from poultry (Rahman et al. 1997; Rajkhowa et al. 2018), has been reported.

The emergence and spread of antimicrobial resistance among zoonotic *Salmonella* has become a public health threat. *Salmonella* strains showing “clinically important resistance” to some agents, such as extended-spectrum cephalosporins and fluoroquinolones, have been isolated from livestock (Li et al. 2013). In most developing countries, misuse and overuse of antibiotics have contributed to the increasing trend of multi-resistance in *Salmonella* (Ed-Dra et al. 2017). *Salmonella* with antibiotic resistance in contaminated products could infect humans directly or transmit their resistance genes to human pathogens through the food chain, leading to the failure of antibiotic treatment and posing a serious threat to human health (Nair et al. 2018). This study investigated the occurrence, antimicrobial resistance and virulence gene profiles of *Salmonella* serovars isolated from humans and different species of animals.

Materials and methods

The samples were collected from different parts of the state of Assam, mostly from the districts of Kamrup (metropolitan), Kamrup (rural), Darrang, Morigaon, Sivasagar, Jorhat and Lakhimpur.

Sample collection

A total of 1231 different samples (Supplementary Table 1), consisting of faecal swabs from diarrhoeic (83) and apparently healthy (60) humans, apparently healthy (101) and diseased (165) cattle, diarrhoeic (60) and apparently healthy (30) pigs, diarrhoeic (53) and apparently healthy (50) goats, diarrhoeic (302) and apparently healthy (103) poultry, apparently healthy (208) wild birds, apparently healthy (8) lizards (*Gecko gecko*), apparently healthy (6) tigers (*Panthera tigris*) and mice (*Mus musculus*) (2), were collected from different parts of Assam, India. The type of sample was either faecal matter or part of the intestine, in case the of dead animals. Fresh samples were collected in sterile sample containers or in Cari-Blair transport medium w/o charcoal (Sisco Research Laboratories, Mumbai, India), in case of anticipated delay from collection to processing by not more than 48 h.

Isolation

Immediately after arriving in the laboratory, the samples were placed in pre-enrichment nonselective medium (sterile buffered peptone water broth) and incubated at 37 °C overnight, followed by inoculation of 1 ml overnight broth into 9 ml of selective enrichment (Selenite broth or Rappaport Vassiliadis soy peptone, HiMedia, Mumbai, India) broth and incubation at 37 °C overnight. Both broth cultures were kept at constant shaking at 250 rpm in a shaking incubator. The overnight turbid broth was inoculated to Brilliant Green Agar (BGA; HiMedia), and suspected positive colonies were picked and streaked on MacConkey's Lactose Agar (MLA; HiMedia). Both plates were incubated overnight at 37 °C. For the preliminary identification of *Salmonella*, the suspected cultures were subjected to biochemical tests, viz. indole, urease and H₂S production, as well as the production of lysine and ornithine decarboxylase. Additionally, growth characteristics on triple sulfate iron (TSI) agar (HiMedia) slants were also studied for the determination of the K/A (acid/alkali production) reaction.

Molecular detection of *Salmonella*

Reference strains of *Salmonella* used in this study (supplementary Table 2) were obtained from the microbial type culture center (MTCC), Chandigarh, American type culture center, USA, and the National Institute of cholera and enteric diseases, ICMR, (NICED), Kolkata. The isolates were grown in Luria Bertani (LB) broth (HiMedia) overnight at 37 °C, and 2 ml of the overnight cultures were centrifuged at 10,000 x g for 10 min. The supernatants were discarded, and the pellets were suspended in a total volume of 100 µl of 1X TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0) and boiled at 100 °C for 10 min. After boiling, the cell suspensions were immediately cooled on ice for 10 min and centrifuged at 15,000 x g for 10 min at 4 °C. the supernatant was then collected without disturbing the sediment. Subsequently, DNA quantification was performed using a NanoDrop1000 spectrophotometer (Thermo fisher scientific, USA) and stored at –20 °C for future use as template DNA for PCR (Choudhury et al. 2016)

The isolates were confirmed as *Salmonella* by detecting the specific gene *invA* by simplex PCR. A reaction mixture of 25 µl was prepared with 12.5 µl of Dream Taq PCR master mix (Fermentus, USA), 0.5 µl of each primer (10 pmol/µl), 1.0 µl of template DNA (100 ng) and 10.5 µl of nuclease-free water. The PCR condition was standardised at 94 °C for 5 min, followed by 30 cycles at 94 °C for 30 s, at 56 °C for 1 min and at 72 °C for 1 min; the final extension step consisted of 72 °C for 5 min, followed by an infinite halt at 4 °C. The amplified products were run on 1.5% agarose gel and visualised and documented in Gel Doc XR⁺ (Bio-Rad, USA).

Serotyping

Salmonella isolates were sent for serotyping at the National Institute of Cholera and Enteric Diseases (NICED), Kolkata, India, and the National *Salmonella* and *Escherichia* Center (NSEC), Central Research Institute, Kasauli, Himachal Pradesh, India.

Antimicrobial susceptibility test and determination of minimum inhibitory concentration (MIC)

Antimicrobial susceptibility and MIC values were determined by the disk diffusion method and the E-test, using 6-mm antibiotic disks (HiMedia) and Ezy MICTM (HiMedia) strips, respectively, according to the Clinical and Laboratory Standards Institute guidelines (CLSI Guideline 2019). Pure colonies from agar plates were inoculated in Luria Bertani broth for 6–8 h to obtain log-phase growth. Following incubation, turbidity was adjusted to match that of a 0.5

McFarland standard. The containers with the disks and strips were removed from the –20 °C to equilibrate to room temperature for 30 min. Sterile 2% Muller Hinton Agar (Sisco Research Laboratories) plates were inoculated by evenly swabbing with a sterile cotton swab (HiMedia). Antibacterial disks and Ezy MICTM strips were placed on the surface aseptically as recommended by the manufacturers and incubated at 37 °C overnight. Inhibition zones and break-point values were recorded after overnight incubation. In total, nine antimicrobial agents were used for both the tests, comprising eight different groups (Supplementary Table 3). Ampicillin (AMP), tetracycline (TET), cefotaxime (CTX), ciprofloxacin (CIP), gentamicin (GEN), chloramphenicol (CHL), cefepime (CPM) and ceftriaxone (CTR) were common for both tests, whereas trimethoprim (TR), used in the disk diffusion method, was replaced by co-trimoxazole (COT) in the E-test (Supplementary Table 4). The MDR and XDR experiments were performed as described elsewhere (Magiorakos et al. 2012; Schwarz et al. 2010).

Resistance and virulence gene screening

Salmonella isolates exhibiting resistance to different antimicrobial agents were then subjected to PCR for detecting the resistance genes for the antibiotics under consideration. Resistance genes *bla*_{OXA}, *bla*_{TEM}, *bla*_{PSE-1}, *dfrA1*, *dfrA12*, *tetA*, *tetB* and *tetG* were detected using the reported primers and PCR conditions (Supplementary Table 5). The PCR reaction mixture constituents were as mentioned earlier for *invA* gene detection. All *Salmonella* isolates were screened for the presence of 11 virulence genes, with *invA* being the universal detection marker for all isolates. The primer sets used for the PCR amplification of *sipA*, *sipB*, *sipC*, *stn*, *sopB*, *sopE2*, *pefA*, *sefC*, *fepA* and *pagN*, along with PCR annealing temperatures, are listed in Supplementary Table 5. The PCR reaction mixture constituents were as mentioned earlier. The amplified products were run on 1.5% agarose gel and visualised under Gel Doc XR⁺ (Bio-Rad).

Results

Isolation and identification

Out of 1231 samples examined, *Salmonella* was recovered from 88 (7.15%) samples. Among the 663 clinical samples collected from humans and animals with a history of diarrhoea, 68 (10.26%) were positive for *Salmonella*, whereas out of 568 samples collected from apparently healthy animals, 20 (3.52%) were positive for *Salmonella* (Table 1). All 88 *Salmonella* isolates recovered from different sources fermented glucose, mannitol and dulcitol but not lactose and sucrose. All isolates were positive for the methyl red test,

Table 1 Isolation of *Salmonella* from human and different species of animals and birds

Host	No. of samples examined			No. of isolates recovered	
	Total	Diarrhoeic	Apparently Healthy	Diarrhoeic	Apparently Healthy
Poultry	405	302	103	17 (5.63%)	7 (6.8%)
Cattle	266	165	101	18 (10.91%)	–
Wild Birds	208	–	208	–	10 (4.81%)
Human	143	83	60	28 (33.73%)	–
Goat	103	53	50	–	–
Pig	90	60	30	5 (8.3%)	–
Gecko	8	–	8	–	1 (12.5%)
Tiger	6	–	6	–	1 (16.67%)
Mice	2	–	2	–	1 (50%)
Total	1231	663	568	68 (10.26%)	20 (3.52%)

citrate usage and hydrogen sulphide production and negative for indole production, the Voges-Proskauer (VP) test and urease production. The isolates showed a yellow butt, black middle and pink top on inoculation into Triple Sugar Iron (TSI) Agar slant.

Serotyping of *Salmonella*

Out of the 88 isolates, 22 (25%) belonged to *S. enterica* subsp. *enterica* serovar Enteritidis, 21 (23.86%) to serovar *S. Weltevreden*, 16 (18.20%) to *S. Typhi* and 14 (15.90%) to *S. Newport*, whereas 7 (7.95%) isolates were found to be untypable. Detailed results of serotyping of the *Salmonella* strains are presented in Table 2. In the case of human samples, *S. Typhi* (62.50%) was the most frequently isolated serovar, followed by serovars *S. Weltevreden* and *S. Newport*. However, *S. Newport* (52.63%) was found the most abundant serovar in cattle, followed by *S. Weltevreden* and *S. Enteritidis*.

Antimicrobial resistance detected by the disk diffusion test

All 88 *Salmonella* isolates recovered from different sources in this study were subjected to antimicrobial susceptibility tests by the disk diffusion method. Of the 88 isolates, 60 (68.18%) were resistant to cefotaxime, 58 (65.91%) to gentamicin, 54 (61.36%) to tetracycline, 43 (48.86%) to trimethoprim, 40 (45.45%) showed resistance to ampicillin, 10 (11.36%) to ceftriaxone, 9 (10.23%) to chloramphenicol and 7 (7.95%) each to ciprofloxacin and cefepime, irrespective of serovars. Details of antibiotic resistance are shown in Table 3. Ciprofloxacin resistance was shown by *S. Weltevreden* from poultry, *S. Typhi* from humans, *S. Enteritidis* from wild birds and poultry, *S. Typhimurium* from humans and untypable isolates from pigs and wild birds. None of the isolates of serovars *S. Litchfield*, *S. Newport*, *S. Kentucky*, *S. Idikan*, *S. Paratyphi B* and *S. Virchow* showed resistance to ciprofloxacin.

Table 2 Distribution of various serovars among *Salmonella* isolates recovered from different sources

Sl. No.	Serotype	Antigenic Structure	No. of isolates	Source/ Host
1	<i>Salmonella</i> Enteritidis	9,12:g,m	22	Cattle (2), Pig (1), Human (2), Poultry (13), Wild Birds (4)
2	<i>Salmonella</i> Weltevreden	3{10},r,z ₆	21	Poultry (5), Human (4), Cattle (7), Pig (1), Wild Birds (3), <i>Gecko gecko</i> (1)
3	<i>Salmonella</i> Typhi	9,12,Vi,d	16	Human
4	<i>Salmonella</i> Newport	6, 8,e, h, 1, 2	14	Cattle (9), Human (2), Poultry (3)
5	<i>Salmonella</i> Typhimurium	1,4,5,12,i,1,2	2	Human
6	<i>Salmonella</i> Litchfield	8,6,8,l,v,1,2	2	Poultry
7	<i>Salmonella</i> Kentucky	8,8,20,I,z ₆	1	Tiger
8	<i>Salmonella</i> Idikan	13,1,3,23,i,1,5	1	Human
9	<i>Salmonella</i> Paratyphi B	1,4,5,12,b,1,2	1	Human
10	<i>Salmonella</i> Virchow	7,6,7,14,r,1,2	1	Mouse
11	Untypable	–	7	Poultry (1), Pig (3), Wild Birds (3)

Table 3 Absolute (n) and relative frequencies (%) of *Salmonella* isolates from different host species showing resistance against different antimicrobial agents

Serovars	Host	No. of isolates	Ampicillin (10 mcg)	Tetracycline (10 mcg)	Cefotaxime (30 mcg)	Ciprofloxacin (5 mcg)	Gentamicin (30 mcg)	Chloramphenicol (25 mcg)	Cefepime (30mcg)	Ceftriaxone (30 mcg)	Trimethoprim (25 mcg)
<i>S. Enteritidis</i>	Wild bird	4	2 (50)	1 (25)	2 (50)	2 (50)	0	0	1 (25)	1 (25)	1 (25)
	Poultry	14	8 (57.14)	7 (50)	9 (64.28)	1 (7.14)	8 (57.14)	0	0	0	8 (57.14)
	Cattle	2	0	2 (100)	2 (100)	0	2 (100)	0	0	0	2 (100)
	Pig	1	0	1 (100)	1 (100)	0	1 (100)	0	0	0	1 (100)
	Human	1	0	1 (100)	1 (100)	0	0	0	0	0	1 (100)
<i>S. Weltevreden</i>	Subtotal	22	10 (45.45)	12 (54.55)	15 (68.18)	1 (4.54)	14 (63.64)	0	1 (4.54)	1 (4.54)	13 (59.09)
	Poultry	5	2 (40)	3 (60)	5 (100)	1 (20)	4 (80)	1 (20)	0	1 (20)	2 (40)
	Human	4	1 (28.57)	4 (100)	4 (100)	0	2 (50)	0	0	0	4 (100)
	Cattle	7	2 (28.57)	4 (57.14)	2 (28.57)	0	3 (42.85)	0	1 (14.28)	1 (14.28)	3 (42.85)
	Pig	1	0	1 (100)	0	0	0	1 (100)	1 (100)	0	0
<i>S. Typhi</i>	<i>Gecko</i>	1	1 (100)	0	1 (100)	0	0	0	0	0	1 (100)
	Subtotal	21	7 (33.34)	12 (57.14)	14 (66.67)	1 (4.76)	10 (47.61)	2 (9.52)	2 (9.52)	2 (9.52)	10 (47.61)
	Human	16	5 (31.25)	13 (81.25)	11 (68.75)	3 (18.75)	10 (62.5)	1 (6.25)	1 (6.25)	3 (18.75)	8 (50)
	Poultry	3	1 (33.33)	1 (33.33)	1 (33.33)	0	2 (66.67)	1 (33.33)	1 (33.33)	1 (33.33)	1 (33.33)
	Cattle	9	7 (77.78)	5 (55.55)	4 (44.44)	0	7 (77.78)	1 (11.11)	0	0	3 (33.33)
<i>S. Litchfield</i>	Human	2	1 (50)	1 (50)	2 (100)	0	2 (100)	1 (50)	1 (50)	2 (100)	1 (50)
	Subtotal	14	9 (64.28)	7 (50)	7 (50)	0	11 (78.57)	3 (21.42)	2 (14.28)	3 (21.42)	5 (35.71)
	Poultry	2	1 (50)	1 (50)	2 (100)	0	2 (100)	0	0	0	1 (50)
	Wild bird	3	1 (33.33)	0	2 (66.67)	0	1 (33.33)	0	0	0	0
	Subtotal	2	1 (50)	1 (50)	2 (100)	0	2 (100)	0	0	0	1 (50.00)
<i>S. Typhimurium</i>	Human	2	1 (50)	2 (100)	1 (50)	1 (50)	1 (50)	0	0	0	1 (50)
	Human	1	1 (100)	1 (100)	1 (100)	0	1 (100)	1 (100)	0	0	1 (100)
	Tiger	1	1 (100)	1 (100)	1 (100)	0	1 (100)	0	0	0	0
	Human	1	0	1 (100)	1 (100)	0	1 (100)	0	0	0	1 (100)
	Mouse	1	0	1 (100)	0	0	1 (100)	1 (100)	0	0	0
Untypable	Poultry	1	1 (100)	1 (100)	1 (100)	0	1 (100)	0	0	0	0
	Pig	3	1 (33.33)	1 (33.33)	3 (100)	3 (100)	2 (66.67)	1 (33.33)	0	1 (33.33)	2 (66.67)
	Wild bird	3	3 (100)	1 (33.33)	3 (100)	1 (33.33)	3 (100)	0	1 (33.33)	1 (33.33)	1 (33.33)
	Subtotal	7	5 (71.43)	3 (42.85)	7 (100)	1 (14.28)	6 (85.71)	1 (14.28)	1 (14.28)	1 (14.28)	3 (42.85)
	Total	88	40 (45.45)	54 (61.36)	60 (68.18)	7 (7.95)	58 (65.91)	9 (10.23)	7 (7.95)	10 (11.36)	43 (48.86)

A total of 34 (38.64%) isolates were found to be multi-drug-resistant (MDR), showing resistance to three or more antimicrobial classes. *Salmonella* Weltevreden isolates showed resistance to cefotaxime (92.31%) and tetracycline (69.23%). All human isolates of serovar *S. Weltevreden* showed resistance to cefotaxime, trimethoprim and tetracycline but not to ceftriaxone and cefepime. Among the poultry isolates, 100% resistance was observed against cefotaxime, whereas no resistance was shown against cefepime. *Salmonella* Newport isolates from humans showed 100% resistance to gentamicin, cefotaxime and ceftriaxone. The serovar *S. Enteritidis* was 100% resistant to cefotaxime, trimethoprim and tetracycline regarding human, pig and cattle isolates, whereas cattle and pig isolates showed 100% resistance to gentamicin. The serovar *S. Typhi*, which is a human host-specific serovar, showed 81.25% tetracycline resistance, whereas resistance to cefotaxime and gentamicin was found 68.75 and 62.5%, respectively.

MIC determined by the E-test

Most of the isolates had MIC values of ≤ 0.125 $\mu\text{g/ml}$ for cefepime, cefotaxime, ciprofloxacin, ceftriaxone and co-trimoxazole, followed by an MIC value of 0.25 $\mu\text{g/ml}$ for both ampicillin and tetracycline. The MIC values of *S. Newport* for all antimicrobial agents ranged between 1 and 16 $\mu\text{g/ml}$; for *S. Typhi*, the values were between 0.5 and 16 $\mu\text{g/ml}$, and for *S. Weltevreden*, they were between 0.25 and 16 $\mu\text{g/ml}$, with only three isolates showing MIC values greater than 256 $\mu\text{g/ml}$. For *S. Enteritidis*, ampicillin and tetracycline MIC values mostly ranged between 0.5 and 8 $\mu\text{g/ml}$, with three isolates showing MIC values greater than 256 $\mu\text{g/ml}$. Chloramphenicol and gentamicin MIC values for *S. Newport*, *S. Weltevreden* and *S. Enteritidis* were between 0.5 and 4 $\mu\text{g/ml}$, whereas the corresponding values were between 0.5 and 8 $\mu\text{g/ml}$ for *S. Typhi* isolates. For *S. Typhimurium*, the MIC values ranged between ≤ 0.125 and 4 $\mu\text{g/ml}$ for cefepime, cefotaxime,

ciprofloxacin, ceftriaxone and co-trimoxazole, whereas for the remaining drugs, the values varied from 4 to 32 $\mu\text{g/ml}$ (Table 4).

Detection of resistance genes

For all samples, DNA extraction was carried out (Choudhury et al. 2016) using a standard protocol; the concentrations ranged between 300 and 500 ng/ μl . *Salmonella* isolates showing resistance to ampicillin, tetracycline and trimethoprim were further subjected to PCR for the detection of resistance genes (*bla*_{TEM}, *bla*_{PSE} and *bla*_{OXA}; *tetA*, *tetB*, and *tetG*; *drfA1* and *drfA12*) corresponding to their phenotypic resistance patterns. Of the 41 isolates showing resistance to ampicillin, 2 (4.88%) carried the *bla*_{OXA} gene, 1 (2.44%) carried *bla*_{PSE} and 28 (63.64%) carried the *bla*_{TEM} gene. Two isolates of the serovar *S. Enteritidis* from poultry carried *bla*_{OXA} and one isolate of *S. Typhi* from humans carried *bla*_{PSE}. The two *S. Newport* isolates from humans and poultry did not possess the *bla*_{TEM} gene. Similarly, three *S. Weltevreden* isolates from poultry and one isolate from wild birds and *Gecko gecko*, belonging to the same serovar, did not possess the *bla*_{TEM} gene. All isolates (100%) of *S. Litchfield*, *S. Typhi*, *S. Typhimurium*, *S. Kentucky* and *S. Idikan* and the untypable isolates carried the *bla*_{TEM} gene.

Of the 54 isolates showing resistance to tetracycline, 9 (16.67%) carried the *tetA* gene and 1 (1.85%) carried the *tetG* gene, whereas none carried the *tetB* gene. Only one untypable isolate from poultry carried the *tetG* gene. However, none of the isolates of *S. Newport*, *S. Kentucky*, *S. Idikan*, *S. Paratyphi B* and *S. Virchow* carried the *tetA* gene. Of the 44 isolates showing resistance to trimethoprim, 5 (11.36%) carried the resistance gene *drfA1*, but none carried *drfA12*. The *drfA1* gene was detected in two isolates of each *S. Typhi* and *S. Enteritidis* and one isolate of *S. Typhimurium* (Table 5).

Table 4 Antimicrobial resistance phenotypes of *Salmonella* isolates ($n=88$)

Antibiotics	Minimum Inhibitory Concentration (MIC) Distribution ($\mu\text{g/ml}$) of 88 <i>Salmonella</i> isolates											
	≤ 0.125	0.25	0.5	1	2	4	8	16	32	64	128	≥ 256
AMP	0	1	5	23	27	7	6	5	2	1	0	11
TET	0	0	23	33	12	4	7	4	2	1	0	2
CPM	64	15	0	3	1	0	1	0	0	0	0	4
CTX	65	19	0	0	0	0	0	0	0	2	1	1
CIP	65	12	1	0	0	2	2	1	5	0	0	0
CHL	0	0	1	1	49	36	1	0	0	0	0	0
GEN	1	2	22	51	6	1	5	0	0	0	0	0
CTR	83	0	1	0	0	0	0	0	4	0	0	0
COT	85	3	0	0	0	0	0	0	0	0	0	0

Table 5 Screening of *Salmonella* isolates from different sources for resistance genes against different antimicrobial agents

Serovars	No. of isolates showing resistance	No. of isolates showing positive for ampicillin resistant genes	Serovars	No. of isolates showing resistance	No. of isolates showing positive for tetracycline resistance genes			Serovars	No. of isolates showing resistance	No. positive for trimethoprim resistance genes
					<i>blaOXA</i>	<i>blaPSE</i>	<i>blaTEM</i>			
<i>S. Litchfield</i>	1	0	0	1 (100)	0	0	<i>S. Litchfield</i>	1	0	0
<i>S. Newport</i>	9	0	0	7 (77.78)	0	0	<i>S. Newport</i>	7	0	0
<i>S. Enteritidis</i>	11	2 (18.18)	0	6 (54.54)	0	0	<i>S. Enteritidis</i>	12	2 (16.67)	13
<i>S. Typhi</i>	4	0	1 (25)	4 (100)	0	0	<i>S. Typhi</i>	13	1 (7.69)	8
<i>S. Weltevreden</i>	8	0	0	2 (25)	0	0	<i>S. Weltevreden</i>	12	2 (16.67)	1
Untypable	5	0	0	5 (100)	0	0	Untypable	3	2 (16.67)	1
<i>S. Kentucky</i>	1	0	0	1 (100)	0	0	<i>S. Kentucky</i>	1	0	0
<i>S. Typhimurium</i>	1	0	0	1 (100)	0	0	<i>S. Typhimurium</i>	2	1 (50)	0
<i>S. Idikan</i>	1	0	0	1 (100)	0	0	<i>S. Idikan</i>	1	0	0
							<i>S. Paratyphi B</i>	1	0	0
							<i>S. Virchow</i>	1	0	0
41		2 (4.88)	1 (2.44)	28 (63.64)	9 (16.67)	0		54	1 (1.85)	44
								5		11.36

Table 6 Distribution of virulence genes in *Salmonella* isolates of different serovars isolated from various sources

Serovar	No. of isolates	Genes											
		<i>inv</i> A	<i>stx</i>	<i>t2544</i>	<i>fep</i> A	<i>pef</i> A	<i>sop</i> E	<i>sop</i> B	<i>sef</i> C	<i>sip</i> A	<i>sip</i> B	<i>sip</i> C	
<i>S. Enteritidis</i>	22	22 (100%)	22 (100%)	22 (100%)	12 (54.55%)	14 (63.64%)	11 (50%)	20 (90.9%)	15 (68.18%)	22 (100%)	22 (100%)	22 (100%)	
<i>S. Weltevreden</i>	21	21 (100%)	21 (100%)	21 (100%)	14 (66.67%)	18 (85.7%)	12 (57.1%)	17 (80.95%)	12 (57.1%)	21 (100%)	21 (100%)	21 (100%)	
<i>S. Typhi</i>	16	16 (100%)	16 (100%)	16 (100%)	14 (87.5%)	13 (81.25%)	13 (81.25%)	14 (87.5%)	6 (37.5%)	16 (100%)	16 (100%)	16 (100%)	
<i>S. Newport</i>	14	14 (100%)	14 (100%)	14 (100%)	8 (57.14%)	13 (92.86%)	11 (78.57%)	14 (100%)	5 (35.71%)	14 (100%)	14 (100%)	14 (100%)	
<i>S. Litchfield</i>	2	2 (100%)	2 (100%)	2 (100%)	1 (50%)	1 (50%)	0	1 (50%)	0	2 (100%)	2 (100%)	2 (100%)	
<i>S. Typhimurium</i>	2	2 (100%)	2 (100%)	2 (100%)	2 (100%)	2 (100%)	1 (50%)	2 (100%)	1 (50%)	2 (100%)	2 (100%)	2 (100%)	
<i>S. Idikan</i>	1	1 (100%)	1 (100%)	1 (100%)	1 (100%)	1 (100%)	0	1 (100%)	1 (100%)	1 (100%)	1 (100%)	1 (100%)	
<i>S. Kentucky</i>	1	1 (100%)	1 (100%)	1 (100%)	0	1 (100%)	1 (100%)	1 (100%)	0	1 (100%)	1 (100%)	1 (100%)	
<i>S. Paratyphi B</i>	1	1 (100%)	1 (100%)	1 (100%)	0	1 (100%)	1 (100%)	0	0	1 (100%)	1 (100%)	1 (100%)	
<i>S. Virchow</i>	1	1 (100%)	1 (100%)	1 (100%)	0	0	0	0	0	1 (100%)	1 (100%)	1 (100%)	
Untypable	7	7 (100%)	7 (100%)	7 (100%)	5 (71.43%)	6 (85.71%)	5 (71.43%)	6 (85.71%)	5 (71.43%)	7 (100%)	7 (100%)	7 (100%)	
Total	88	88 (100%)	88 (100%)	88 (100%)	57 (64.77%)	70 (79.54%)	55 (62.5%)	76 (86.36%)	45 (51.14%)	88 (100%)	88 (100%)	88 (100%)	

Virulence gene detection

All 88 *Salmonella* isolates were subjected to simplex PCR for the detection of 11 important virulence genes (Table 6). The different serovars of *Salmonella* showed variability in their virulence gene profiles. While *invA* was used as a marker for the molecular detection of *Salmonella*, the virulence genes *sipA*, *sipB*, *sipC*, *stn* and *pagN* were also detected in all *Salmonella* isolates, irrespective of the serovar, whereas the *fepA* gene was present in 57 (64.77%) isolates belonging to serovars *S. Enteritidis* (12), *S. Weltevreden* (14), *S. Typhi* (14), *S. Newport* (8), *S. Litchfield* (1) and *S. Idikan* (1), *S. Typhimurium* (2) and untypable (5) isolate. Virulence genes *sopB* (86.36%), *sopE2* (62.5%), *pefA* (79.54%) and *sefC* (51.14%) were present in varying percentages among the *Salmonella* serovars. Out of the 88 isolates screened, 11 (12.5%) isolates belonging to serovars *S. Weltevreden* (7) and *S. Typhi* (4) were positive for all 11 genes, along with three other untypable isolates.

The distribution of virulence genes according to the source of recovery of the isolates revealed that *invA*, *sipA*, *sipB*, *sipC*, *stn* and *pagN* genes were present in all isolates, irrespective of their source of origin. The *sefC* gene was present in 45 (51.14%) isolates from humans, cattle, wild birds, pigs, tigers and poultry. The genes *sopB* and *sopE2* were present in 76 (86.36%) and 55 (62.50%) isolates, respectively, whereas *pefA*, *sefC* and *fepA* genes were present in 70 (79.54%), 45 (51.14%) and 57 (64.77%) isolates, respectively, recovered from humans, cattle, wild birds, pigs, tigers, *Gecko gecko* and poultry.

Discussion

Salmonella infections are a major global public health problem. In the last decade, antimicrobial resistance and multi-drug resistance of *Salmonella* spp. have increased to a great extent, especially in developing countries, which is in line with the increased and indiscriminate use of antimicrobial agents for treating human and animal diseases (Threlfall 2002). In this study, an attempt was made to isolate *Salmonella* from faecal and intestinal samples of humans and animals to study the serotype distribution and antimicrobial resistance patterns and to detect important antibiotic resistance genes and *Salmonella*-specific virulence genes. Of 83 diarrheic stool samples collected from humans, 28 (33.73%) were positive for *Salmonella*. Working in the same geographical region, Borah et al. (2017) could recover *Salmonella* from 45.23% of diarrheic stool samples from humans and from 6.34% of bovine faecal samples. Purkayastha et al. (2015) recovered the organism from 46.67% of diarrheic human stool samples. Our study found 18 (10.90%) out of the 165 diarrheic cattle and 5 (8.30%) out of 60 diarrheic

pigs to be positive for *Salmonella*. A previous study (Rahman 2006) reported a 10.63% prevalence of *Salmonella* in diarrheic piglets.

Of the 302 faecal/intestinal samples we collected from diarrheic poultry, 17 (5.63%) yielded *Salmonella*. There have been reports of the recovery of *Salmonella* from as low as 2.90% (Esteban et al. 2008) and as high as 42% (Chen et al. 2020) from different parts of the world. The moderate to high prevalence of *Salmonella* in poultry recorded in this region might be attributed to the favourable climatic conditions, flaws in biosecurity measures adopted in the farms and to the fact that poultry is the major reservoir of *Salmonella* (Sharma and Das 2016). Details of the antigenic structures of the isolates are presented in Table 2. Earlier studies have also reported that *Salmonella* Enteritidis is the most common serotype worldwide (65% of the isolates), followed by *S. Typhimurium* (12%) (Galanis et al. 2006). This was in close agreement with the present findings. In animals, *S. Typhimurium* is the most common serovar recovered in India, followed by *S. Weltevreden* (Kumar et al. 2009). This was also in partial agreement with the present findings. *Salmonella enterica* ser. Typhimurium is assumed to be particularly pathogenic to humans and has been reported worldwide as the second most frequent serovar causing salmonellosis among humans, after *S. enterica* ser. Enteritidis (Murray et al. 2021).

A higher level of resistance shown by *Salmonella* isolates to ampicillin, gentamicin, cefotaxime and tetracycline might be attributed to the frequent and long-term use of these drugs as therapeutic agents both in humans and animals (Nair et al. 2018). Although chloramphenicol has long been used as a drug of choice for the treatment of salmonellosis in humans, interestingly, a lower number of isolates tested in this study exhibited resistance to this drug. This might be attributed to the current trend of prescribing fluoroquinolones, such as ciprofloxacin, more frequently to treat cases of enteric fever in humans (Jin et al. 2019). Comparatively higher percentages of isolates from cattle and poultry exhibited resistance to most of the antimicrobial agents tested. This might be due to the more frequent use of antimicrobial therapy in these two species, compared to other species, to control and prevent infectious diseases. Recently, chloramphenicol, ampicillin and trimethoprim non-susceptible *S. Typhi*, which was also non-susceptible to ceftriaxone, created havoc in Pakistan and Bangladesh (Klemm et al. 2018). In our study, we also recovered a few isolates showing the same resistance pattern, which is alarming. Higher levels of resistance to ampicillin and tetracycline have also been reported previously (Li et al. 2019), corroborating the present findings. *Salmonella* Enteritidis causes gastroenterological disorders in humans, and its antimicrobial resistance is a huge concern. A high multi-drug resistance was observed in 44% of strains isolated from sausages (Ed-Dra et al. 2019). In this study,

ampicillin (45.45%), tetracycline (54.55%), cefotaxime (68.18%), gentamicin (63.34%) and trimethoprim (59.09%) resistance could be found among *S. Enteritidis* isolates. The findings by the AMRSN, showing a nationwide downward trend in India for resistance to ampicillin, chloramphenicol and trimethoprim-sulfamethaxone among *S. Typhi* (Walia et al. 2019), are in agreement with the present findings. *Salmonella* Weltevreden is found equally both in humans and animals and has importance in both as a pathogen. Wang et al. (2019) stated that antibiotic resistance behaves differently in different hosts for the same serovar. In this study, *S. Weltevreden* showed less resistance to ampicillin, ciprofloxacin, chloramphenicol and cephalosporins. This is indeed a good sign for the region as treatment options are still available; this finding is in agreement with the results of Li et al. (2018). *Salmonella* Weltevreden, *S. Typhi*, *S. Newport* and *S. Enteritidis* were the most frequently isolated serovars in this study. Cefotaxime, a third-generation cephalosporin, was less effective against all these isolates. However, ceftriaxone, another third-generation cephalosporin, along with cefepime, a fourth-generation cephalosporin, was effective against most of the isolates, irrespective of the host and serovar. Interestingly, a moderate level of resistance was shown by the isolates to ampicillin. This may be due to the current practice of the limited use of β lactam antibiotics without a β lactamase inhibitor (such as Clavulanic acid) in the treatment regime for human and livestock diseases. Most of the isolates from humans and cattle were non-susceptible to gentamicin, but moderate resistance was observed for isolates from other host species. This may be due to the abundant use of the drug in these two species for treatment. Ciprofloxacin, a fluoroquinolone, has remained effective as less resistance was shown by the isolates of all serovars against this drug. Isolates from cattle, poultry and pigs showed a moderate level of resistance to chloramphenicol, a life-saving ICU-category drug, and low resistance was exhibited by the isolates from humans, particularly of serovars *S. Weltevreden* and *S. Newport*. Resistance to some antimicrobial agents, such as ampicillin, tetracycline and the second-generation cephalosporins, shown by a limited number of isolates from wild animals and birds, could not be conclusively interpreted due to limited data, but this might have been acquired through cross-infection from domestic or stray animals in the fringe areas of the forests. Antimicrobial resistance in *Salmonella* isolates from wild animals and birds has previously been reported (Botti et al. 2013). Proximity to humans is assumed to influence the antibiotic resistance profiles of the gut bacteria of wild animals living in densely populated microbial habitats in which antibiotics select for resistance (Radhouani et al. 2014).

The increase in antimicrobial resistance is a threat to global health. Trimethoprim is commonly used for treating urinary tract infections (UTI) in all parts of the world.

However, soon after the introduction of the drug, trimethoprim resistance has been reported in several species (Skold 2001). In this study, 41, 54 and 44 phenotypically positive *Salmonella* isolates, which showed resistance to ampicillin, tetracycline and trimethoprim, respectively, were subjected to PCR for the detection of resistance genes conferring resistance to these drugs. Of 41 isolates showing resistance to ampicillin, 28 (63.64%) carried the *bla*_{TEM} gene, 2 (4.88%) carried the *bla*_{OXA} gene, and 1 (2.44%) carried the *bla*_{PSE} gene. The presence of *bla*_{TEM} in the isolates non-susceptible to ampicillin, as observed in this study, suggests that ampicillin resistance in the majority of *Salmonella* is mediated by this gene. Of the 54 isolates showing resistance to tetracycline, 9 (16.67%) carried the *tetA* gene and 1 (1.85%) the *tetG* gene, whereas the *tetB* gene was not detected in any isolate. This suggests the involvement of some other resistance determinants in the other isolates that conferred resistance to tetracycline. In a study in Mumbai, Waghmare et al. (2018) observed a significant increase in tetracycline resistance, with the presence of the *tetA* gene in *Salmonella*, indicating selective pressure for adopting resistance to the tetracycline group of antibiotics. Of the 44 isolates showing resistance to trimethoprim, 5 (11.36%) carried the resistance gene *drfA1*, but none carried *drfA12*. Trimethoprim resistance in *Salmonella* is assumed to be mostly mediated by the *drfA1* gene (Brolund et al. 2010).

The disruption of genes in the *Salmonella* Pathogenicity Island (SPI) I of *S. Typhimurium* and *S. Dublin* blocks the secretion of *Salmonella* invasive proteins. The virulence genes of *Salmonella* spp. encoding five different SIPs (*Salmonella* invasion protein), namely *sipA*, *B*, *C*, *D* and *E*, can induce apoptosis in macrophages (Kaur and Jain 2012) and hence may play a vital role in *Salmonella* pathogenesis. Since *stn* is localised and transcribed in juxtaposition to the gene, which encodes the dehydrogenase regulatory protein, a common and related protein among enteric microorganisms, it is anticipated that *stn* determinant might be prevalent among all salmonellae (Prager et al. 1995). This gene in *Salmonella* is one of the chromosomally encoded genes and codes for the production of enterotoxins. Our observations indicate that the *stn* gene is universally present among all *Salmonella* isolates, irrespective of the serovars, which is in agreement with previous findings (Prager et al. 1995; Murugkar et al. 2003; Chakraborty et al. 2020). In this study, the *sopB* gene was detected in 86.36% of the *Salmonella* isolates, similar to the findings of Rahman (2006), who reported that all 50 isolates of *S. enterica*, belonging to 11 serovars, carried the *sopB* gene, irrespective of their serovars and isolation sources. The inositol phosphate phosphatase, SopB, is one of the effector proteins responsible for the induction of macropinocytosis (Garcia-Gil et al. 2018). The *sefC* gene was present in 51.14% of the isolates. The *sefABC* genes make up part of a complex *sef* operon responsible for

the expression and assembly of SEF14 fimbriae (Clouthier et al. 1993). Rearing conditions, such as the type of housing and management practice, as well as the habitat and nature of the animals are important variables that might have influenced the findings of this study. However, as the corresponding data were not collected, such correlations could not be studied.

Conclusions

The higher antibiotic resistance in cattle and poultry recorded in this study might be attributed to the more frequent and rather indiscriminate use of antimicrobial agents to control and prevent infectious diseases. Moreover, the recovered *Salmonella* isolates exhibiting multidrug resistance and multiple virulence genes suggest a possible risk to humans and animals. Therefore, it is important to rationalise the use of antimicrobial agents to prevent the vertical and horizontal transfer of MDR strains across hosts. The presence of multiple virulence genes in different combinations in all field isolates, revealed by this study, indicates their possible role as a pathogen in the host species. However, a molecular-level phylogenetic analysis is needed to establish the inter-serovar and inter-host relationships among the *Salmonella* isolates.

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Declarations

Competing interests The authors declare no competing interests.

Ethics approval For collecting fecal and swab samples from different sources, informed consent was taken from the individuals/owners

of the animals. Tissue samples were collected from dead animals by taking consent from the concerned authority conducting postmortem examination. The ethical approval for the entire work was obtained vide Approval No. 770/ac/CPCSEA/FVSc/AAU/IAEC/16–17/388.

Consent to participate Necessary consent from the concerned authority of each farm and/or individual was taken before collecting the samples.

Consent for publication All the authors have given their consent for the publication of the manuscript in this form.

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